Lecture 11: Node Voltage Analysis

EE201 – Circuit Theory I

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current voltage, node voltage e supernode

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Keep the original circuit

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages **e** as the unknowns, KVL holds automatically.

Only KCL is left to write.

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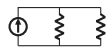
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Later, mesh current analysis will do the same with mesh currents.

Roadmap









Current sources | **Voltage sources** | **Dependent sources** | **The ground**

1 — 2 — 3 — 4

the procedure, on an example

matrix form

one end at the ground

between two nodes: Method I

Method II: the supernode

a constraint equation

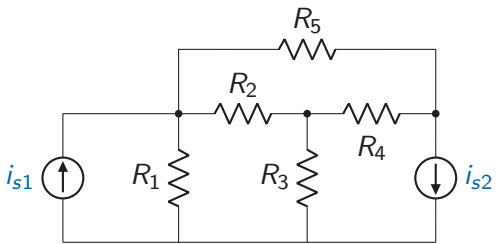
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Each station adds a new kind of source; the last one asks about the ground.

Example 11.1: node voltage analysis step by step

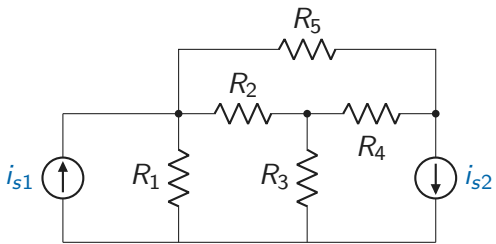
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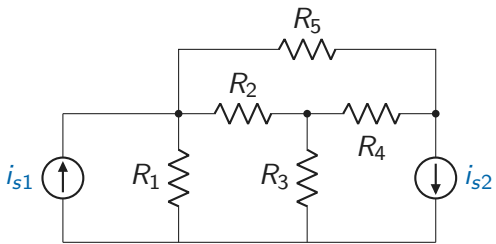


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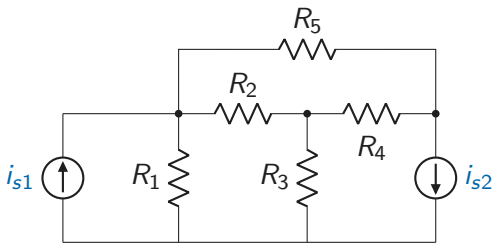
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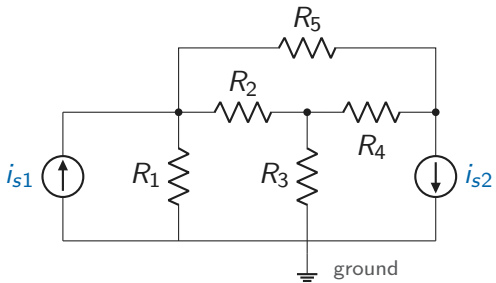
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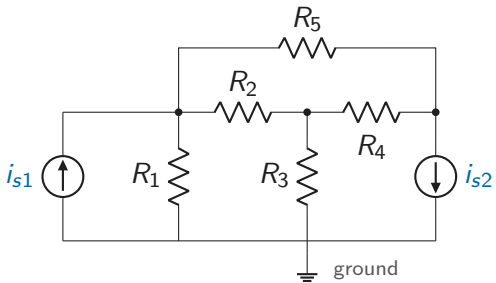
We find them in four steps, on this circuit.

Step 1: choose a reference node



Choose one node as the reference node, the ground.

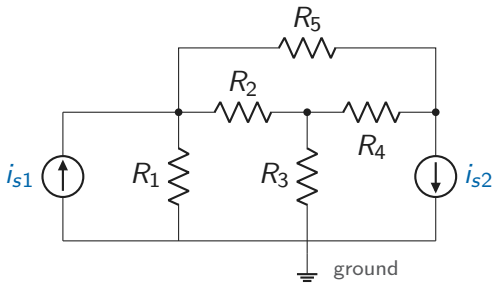
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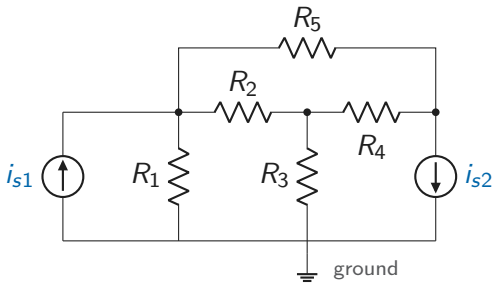


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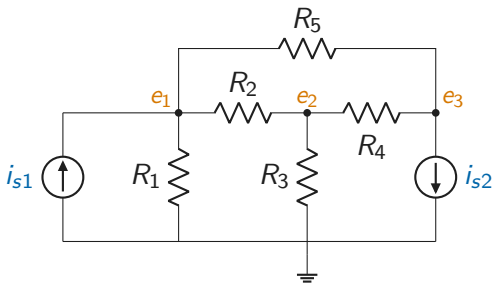
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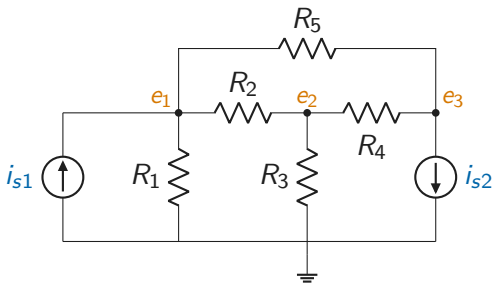
Any node can be the ground; we come back to this at the end.

Step 2: label the node voltages



Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

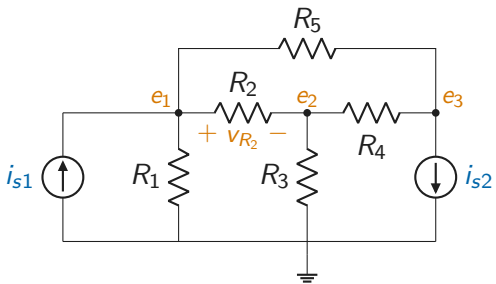
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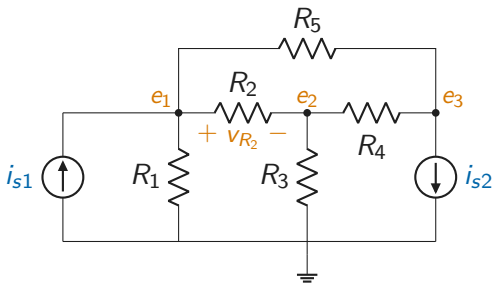


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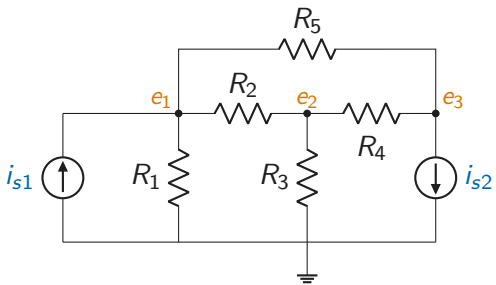
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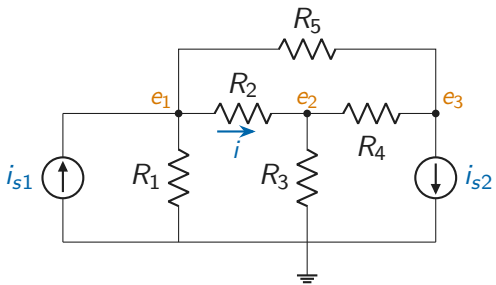
So KVL holds automatically: around R_1, R_2, R_3, $v_{R_1} = v_{R_2} + v_{R_3}$ is $e_1 = (e_1 - e_2) + e_2$.

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Ohm's law gives each resistor current from its two node voltages.

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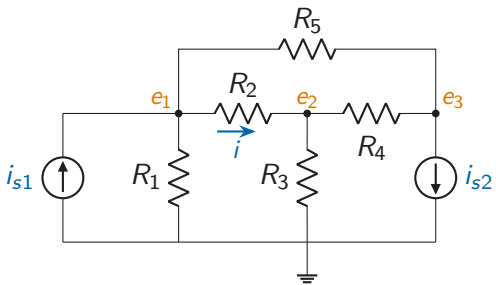


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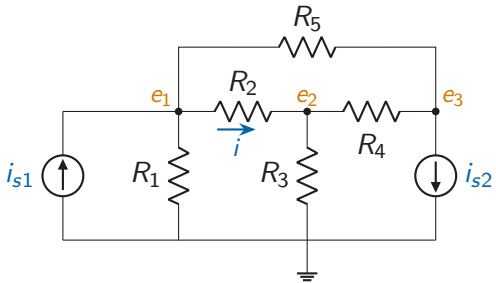
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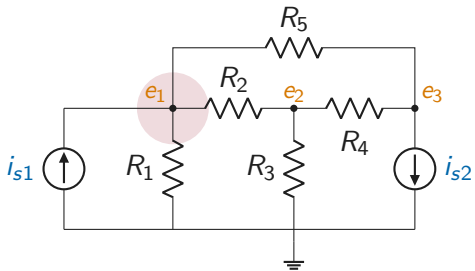
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The rule: own node voltage minus the other one, over R.

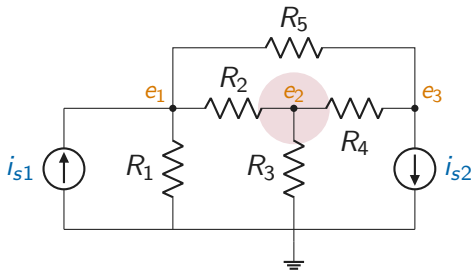
Step 4: KCL at every node except the ground



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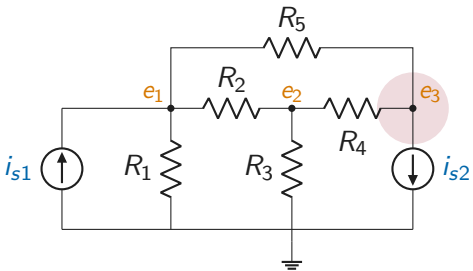
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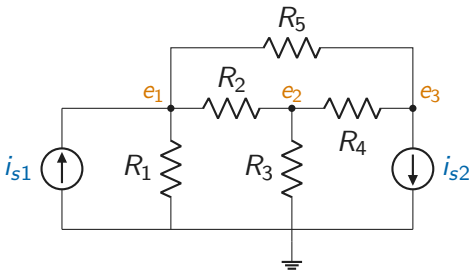
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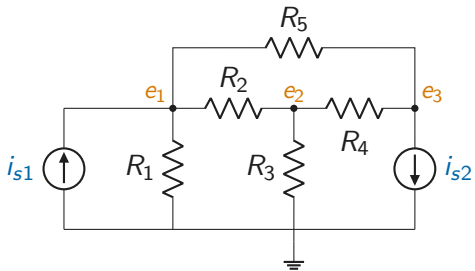
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Three equations, three unknowns.

Then Steps 2 and 3 give every branch voltage and current.

Roadmap



Current sources

1

the procedure, on an example ✓

matrix form



Voltage sources

2

one end at the ground

between two nodes: Method I

Method II: the supernode



Dependent sources

3

a constraint equation



The ground

4

does its choice matter?

Each station adds a new kind of source; the last one asks about the ground.

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

$$\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
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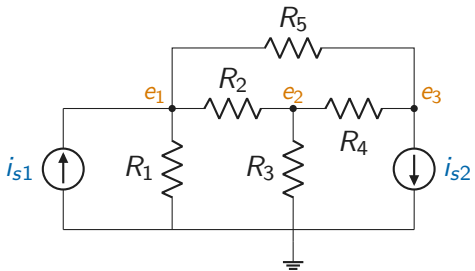
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Y_n is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_s$.

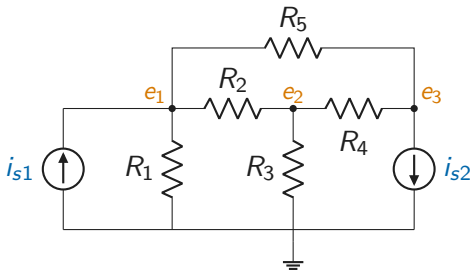
Writing Y_n directly from the circuit



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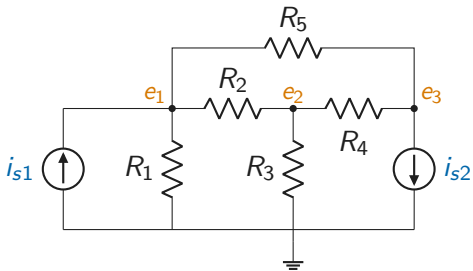


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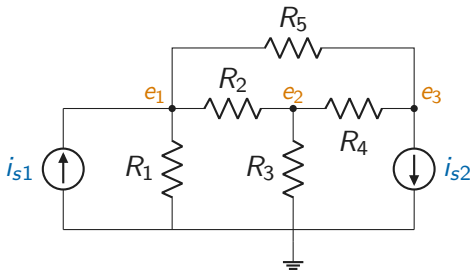
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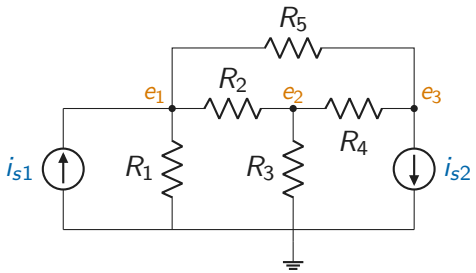
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Right side k: the sum of the source currents entering node k.

So Y_n is symmetric: the entry (k,j) equals the entry (j,k).

Why is Y_n symmetric?

The reduced incidence matrix A

One row for each node except the ground, one column for each branch.

Entry $+1$ if the branch leaves the node, -1 if it enters, 0 otherwise.

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$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

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$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

$$Y_n^T = A_R\, G^T A_R^T = A_R\, G\, A_R^T = Y_n$$

Why is Y_n symmetric?

The reduced incidence matrix A

One row for each node except the ground, one column for each branch.

Entry $+1$ if the branch leaves the node, -1 if it enters, 0 otherwise.

Number the resistors first, then the current sources: $A = [\, A_R \;\; A_S \,]$, $\mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}$.

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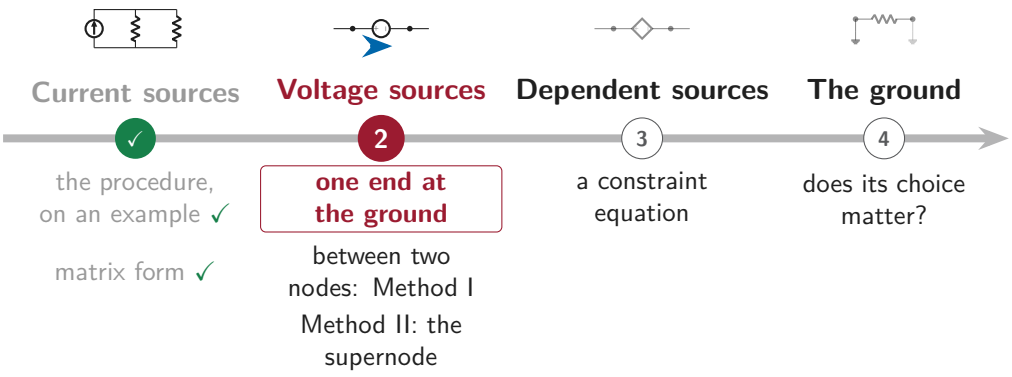
$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

$$Y_n^T = A_R\, G^T A_R^T = A_R\, G\, A_R^T = Y_n$$

G is diagonal, so Y_n is symmetric.

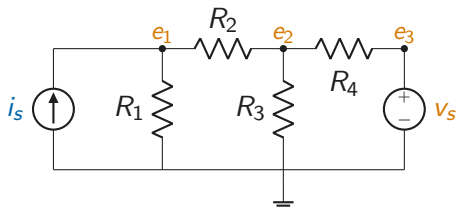
$-A_S \mathbf{i}_S$ adds up the source currents entering each node.

Roadmap



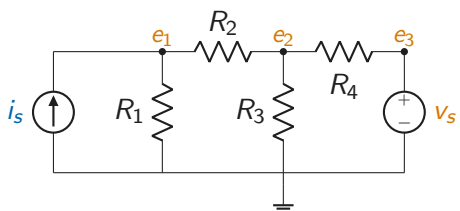
Each station adds a new kind of source; the last one asks about the ground.

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

Example 11.2: a voltage source to the ground

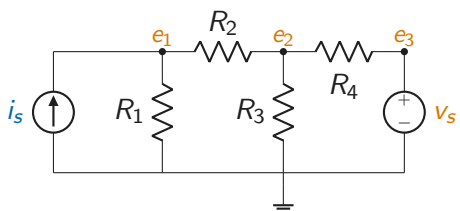


Unknowns e_1, e_2, e_3: three equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Example 11.2: a voltage source to the ground



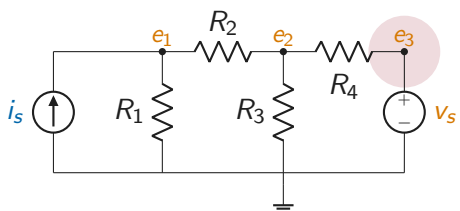
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Question. What does KCL at node 3 give?

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

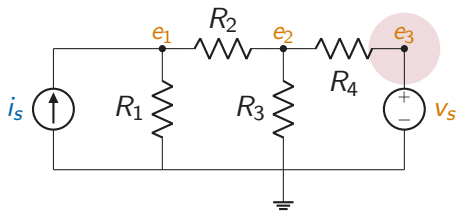
Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Question. What does KCL at node 3 give?

The current through v_s is unknown: KCL gives no useful equation in e_1, e_2, e_3.

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

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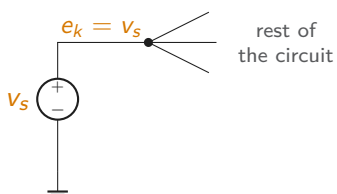
But $e_3 = v_s$: this is the third equation.

A voltage source to the ground

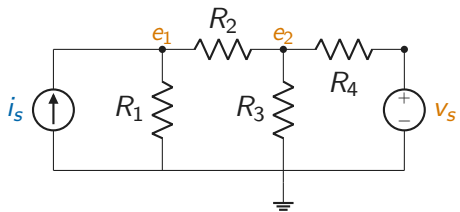
Remark

If an independent voltage source v_s is connected between the ground and node k, then $e_k = v_s$ is known.

So e_k is not an unknown, and we do not write KCL at node k.



Example 11.2, revisited

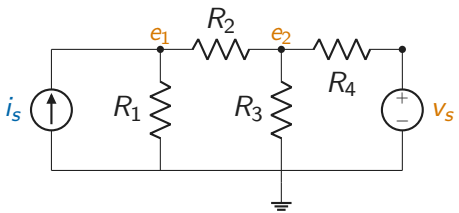


Unknowns e_1, e_2 only: two equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

Example 11.2, revisited



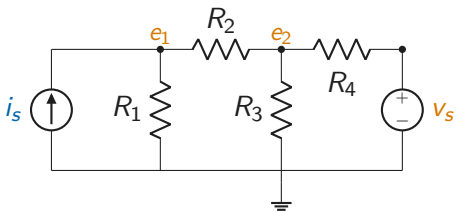
Unknowns e_1, e_2 only: two equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

Example 11.2, revisited



Unknowns e_1, e_2 only: two equations needed.

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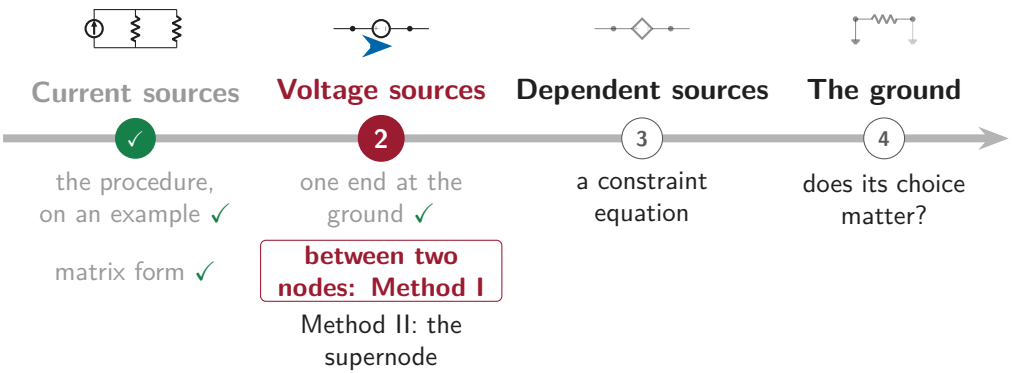
Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

The known v_s moves to the right side.

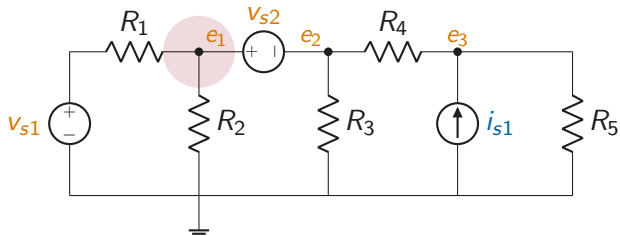
v_s/R_4 acts like a current source into node 2: a source transformation.

Roadmap



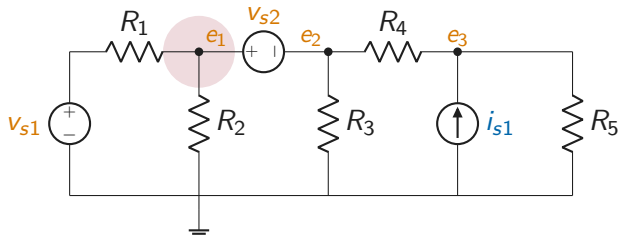
Each station adds a new kind of source; the last one asks about the ground.

Example 11.3: a voltage source between two nodes



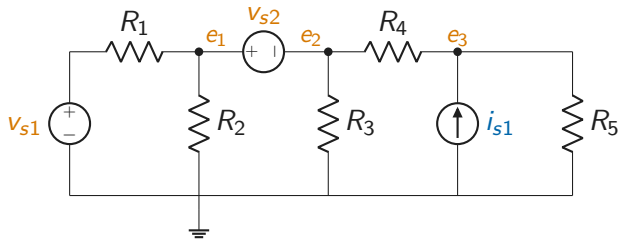
- **Question.** Write KCL at node 1: what is the current through v_{s2}?

Example 11.3: a voltage source between two nodes



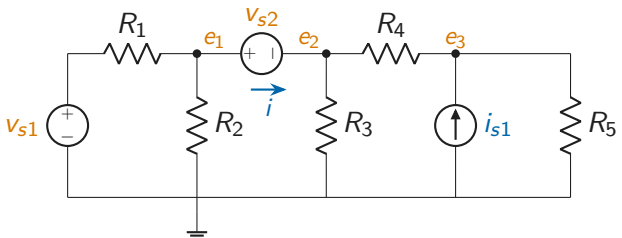
- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.

Example 11.3: a voltage source between two nodes



- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.
- **Modified node voltage analysis** offers two ways out: Method I (its current is a new unknown) and Method II (a supernode).

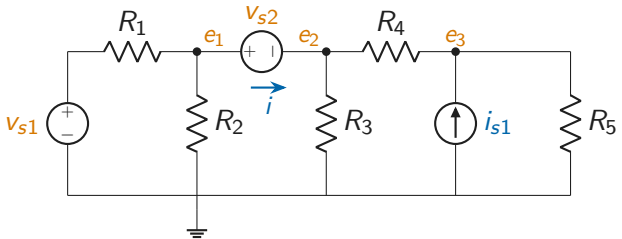
Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown

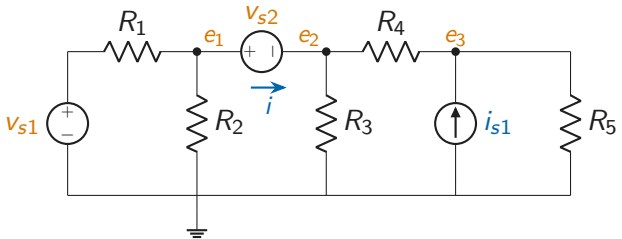


Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown



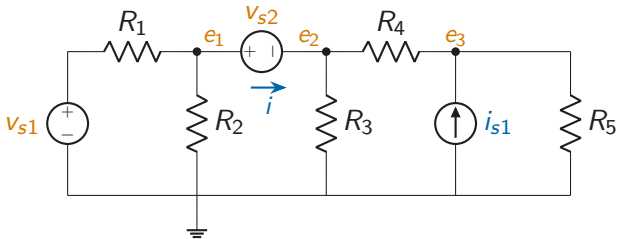
Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

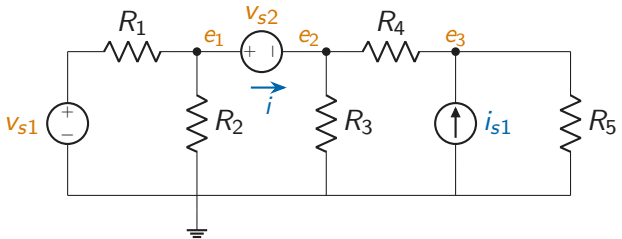
Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Four unknowns, three equations: one more is needed.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Four unknowns, three equations: one more is needed.

Constraint: $e_1 - e_2 = v_{s2}$, the law of the source.

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

- The first three rows are KCL; the last row is the constraint.

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

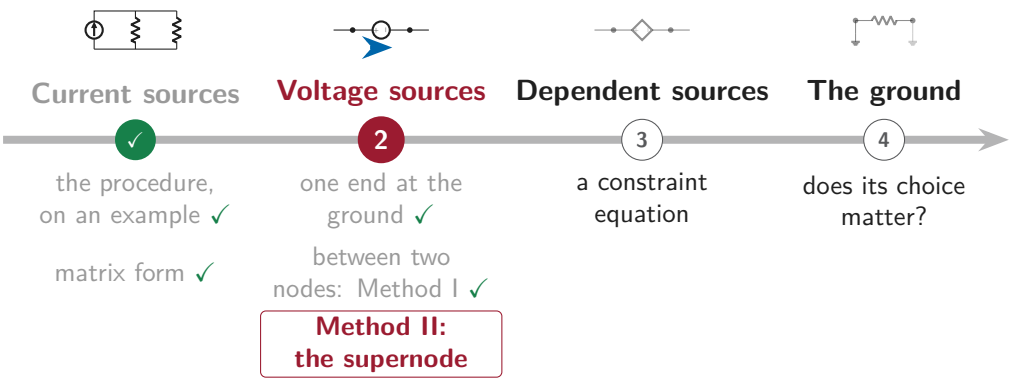
- The first three rows are KCL; the last row is the constraint.
- The unknown current i is part of the solution: we get it directly.

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

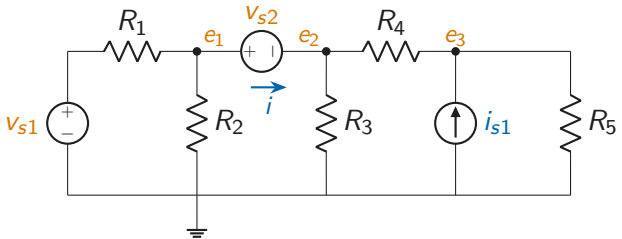
- The first three rows are KCL; the last row is the constraint.
- The unknown current i is part of the solution: we get it directly.
- The price: one more unknown and one more equation.

Roadmap



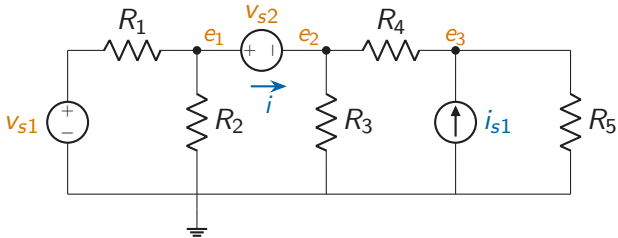
Each station adds a new kind of source; the last one asks about the ground.

Method II: the supernode



Question. Can we get rid of the unknown i?

Method II: the supernode

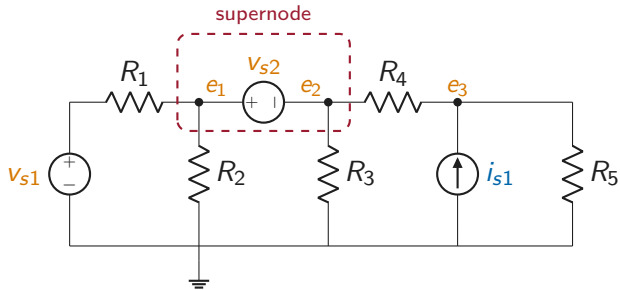


Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

Method II: the supernode



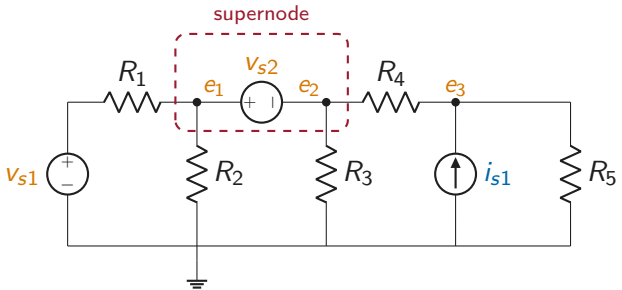
Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Method II: the supernode



Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

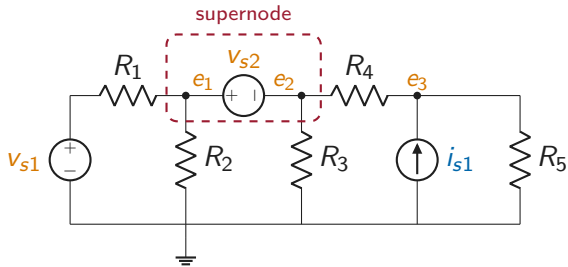
i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Supernode

Write KCL as if the two nodes of the voltage source were a single node.

Method II: the equations



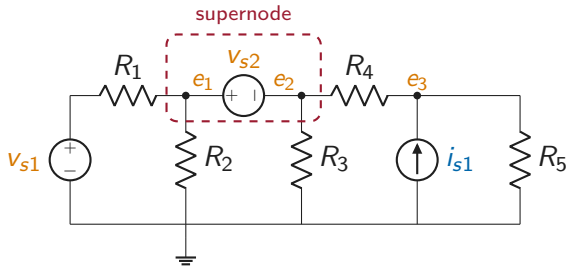
Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

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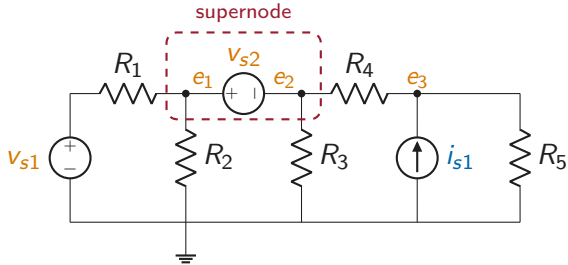
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Three unknowns, three equations.

The matrix is not symmetric any more.

Method I or Method II?

	Method I	**Method II**
extra unknown	the current i of the source	none
KCL	at every non-ground node	at the supernode, and the others
extra equation	the constraint $e_1 - e_2 = v_{s2}$	the same constraint
size (this example)	4×4	3×3
the current i	part of the solution	from KCL afterwards

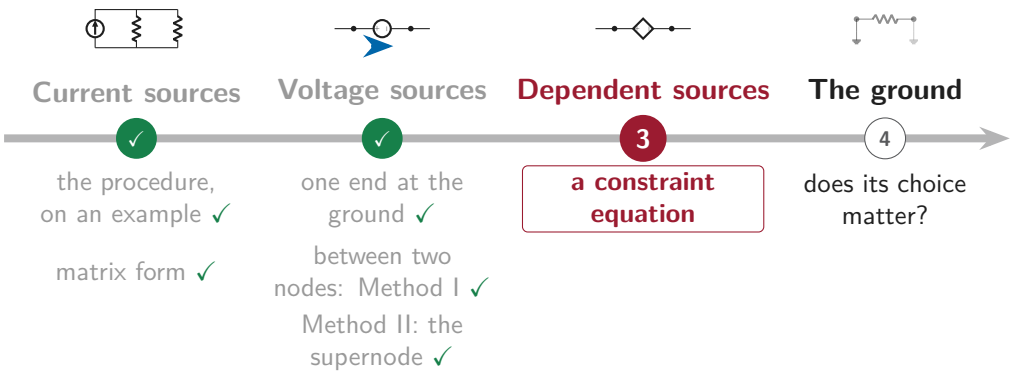
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Both give the same node voltages.

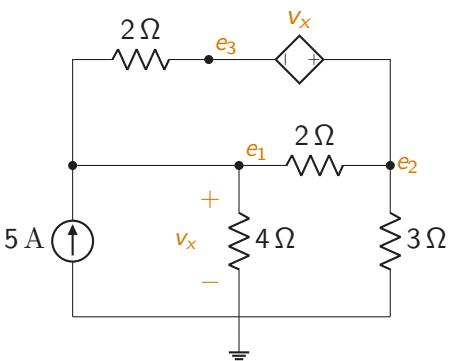
The constraint is always the law of the voltage source.

Roadmap



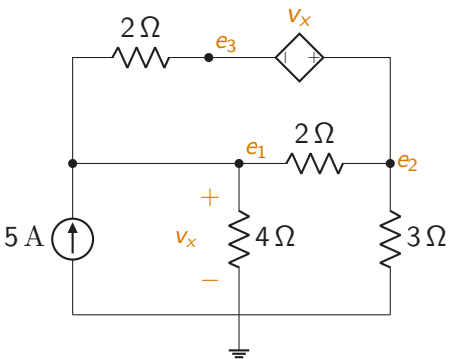
Each station adds a new kind of source; the last one asks about the ground.

Example 11.4: a dependent source



Question. Find the power absorbed by the 3 Ω resistor.

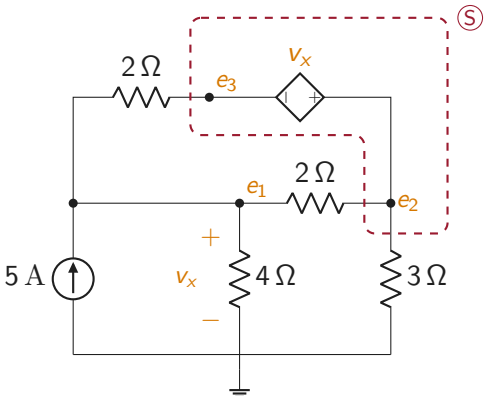
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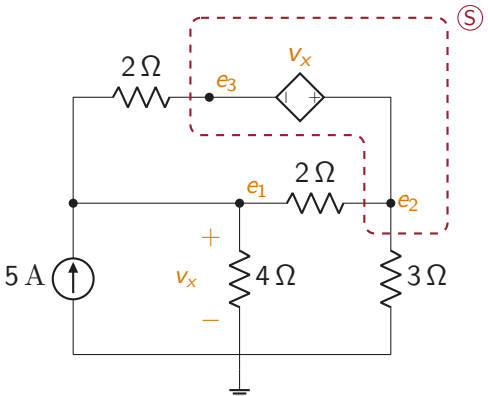


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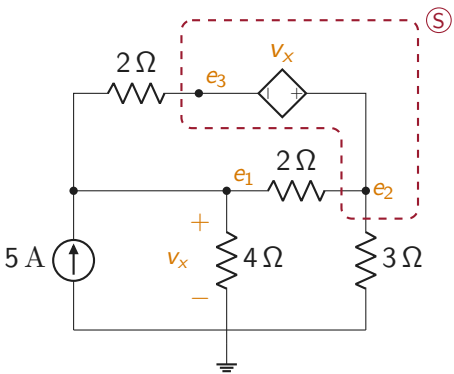
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So use a supernode Ⓢ around nodes 2 and 3.

Unknowns: e_1, e_2, e_3.

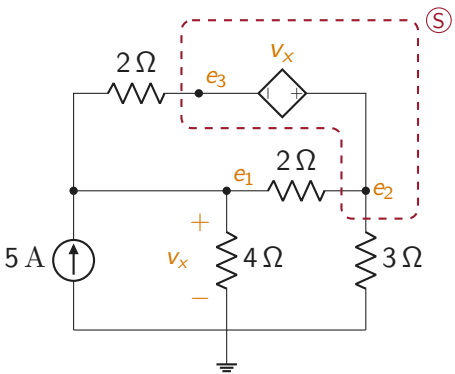
Equations: KCL at node 1, KCL at Ⓢ, and a constraint.

Example 11.4: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

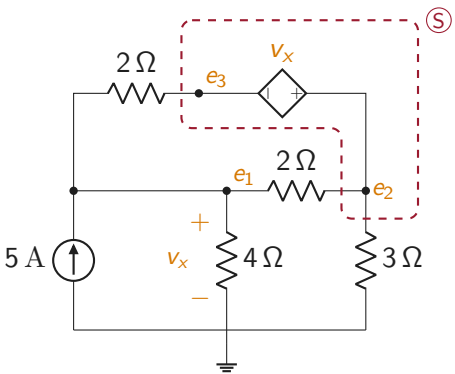
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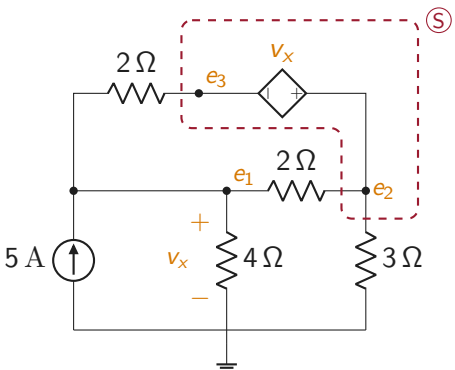


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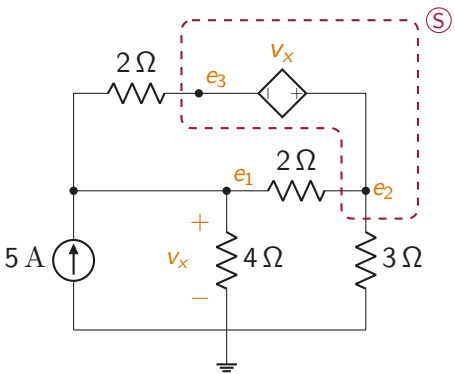
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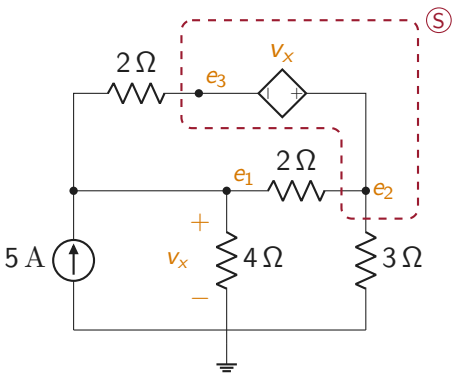
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Three equations, three unknowns.

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)

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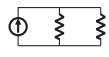
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Remark

If a dependent voltage source sits between node k and the ground, do not write KCL at node k: its current is not known.

Write the constraint that the dependent source gives instead.

Roadmap









Current sources ✓

the procedure, on an example ✓

matrix form ✓

Voltage sources ✓

one end at the ground ✓

between two nodes: Method I ✓

Method II: the supernode ✓

Dependent sources ✓

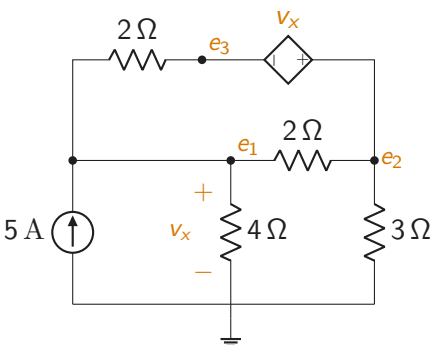
a constraint equation ✓

The ground 4

does its choice matter?

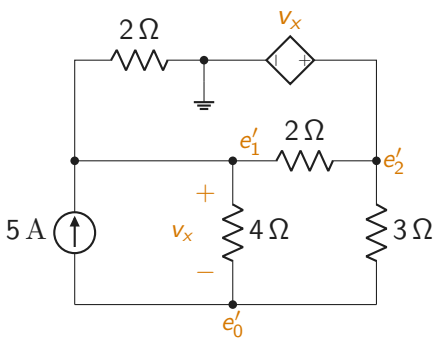
Each station adds a new kind of source; the last one asks about the ground.

Exercise: choose node 3 as the ground



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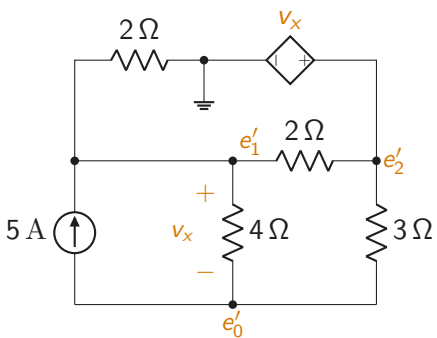
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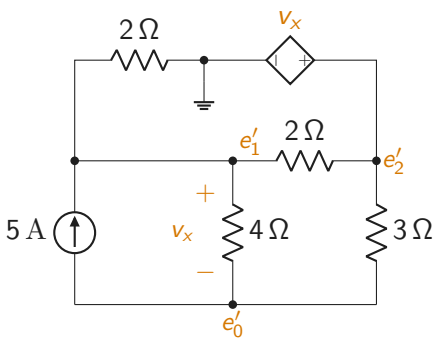


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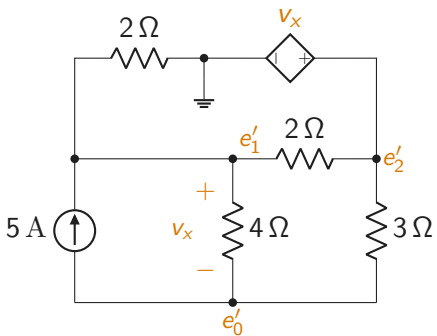
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Constraint: $e_2' = v_x = e_1' - e_0'$; no KCL at node 2.

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- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

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Tip: put the ground at one end of a voltage source; this saves a supernode.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_s$, from three simple rules; Y_n is symmetric
a voltage source from the ground to node k	$e_k = v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages: $\mathbf{v} = A^T\mathbf{e}$, then the currents from the element laws.